PRD: Healthcare App '*Cure*'

1.0 Introduction

1.1 Product Overview

XYZ Ltd. is launching an innovative, comprehensive digital healthcare platform tailored to the unique needs of Bangladesh. The project aims to bridge critical healthcare access gaps for rural and under-served populations by leveraging both digital and offline strategies. By simplifying onboarding, enabling low-bandwidth interactions, integrating familiar payment methods, and addressing affordability challenges, the platform seeks to deliver trusted, continuous care for Bangladeshis regardless of location or digital literacy. The vision is to create a seamless, inclusive healthcare journey that enhances health outcomes and builds deep community trust across the country.

1.2 Product Goals

Goal ID	Goal Description	Metric for Success	Strategic Alignment
G1	Onboard 100,000 rural and non-digital-native users within 12 months	100,000 verified user registrations	Expands market reach and inclusivity
G2	Achieve 80% consultation completion rate through low-bandwidth features and multilingual UI	≥80% consultation completion rate	Improves user engagement and operational efficiency
G3	Integrate MFS (Mobile Financial Services) with ≥90% payment satisfaction from users	≥90% positive payment feedback	Builds user trust and accelerates cashless healthcare adoption
G4	Reduce post-consultation out-of-pocket expenses by offering e-prescription and cashback incentives	20% decrease in reported out-of-pocket spend	Addresses affordability and increases repeat usage

1.3 Product Scope

In Scope:

- Offline marketing and PHC (Primary Healthcare) worker training for onboarding
- Low-bandwidth app download and data-light consultation modes
- Multilingual user interface (Bangla/English)
- Simple symptom checker and digital patient history
- Integration with major Mobile Financial Services (bKash/Nagad)

- Voice and video consultation options
- e-Prescription generation and pharmacy network integration
- MFS-based cashback for pharmacy and hospitalization expenses
- SMS reminders and chronic disease management modules

Out of Scope:

- In-person doctor visits coordination
- AI-based advanced diagnostic tools (Phase 2)
- Hospital/clinic workflow automation
- Full insurance underwriting and claims processing
- Third-party wellness service integrations

1.4 Success Metrics

Metric ID	Key Performance Indicator (KPI)	Target Value	Data Source
M1	New User Registrations (rural/low digital)	>100,000	User analytics
M2	Consultation Completion Rate	≥80%	App usage logs
M3	Average App Load Time (2G network)	<5 seconds	Performance monitoring
M4	e-Prescription Redemption Rate	>70%	Pharmacy API/Reports
M5	Satisfaction with MFS Payments	≥90% positive	In-app user surveys

2.0 Market Analysis

2.1 Target Audience

Primary Audience:

- Rural and peri-urban Bangladeshis, especially non-digital-natives (ages 25–65)
- Low- to middle-income families, often with limited or no health insurance
- Individuals managing chronic conditions or dependent on public health systems
- Rely on community health workers and local pharmacies for health information

Secondary Audience:

- Urban digital-savvy individuals seeking convenient consultations and trustworthy prescriptions
- Primary healthcare workers and local clinic operators

Ideal Customer Profile (ICP):

- Name: Ayesha Begum, 38, Village Homemaker
- Daily responsibilities: Household care, caring for elders/children, managing family health needs
- Challenges: Infrequent internet access, language barriers, distrust of digital health, high out-of-pocket costs
- Motivations: Trusted health advice, affordability, local payment options, ease of use

2.2 Competitive Analysis

Competitor Name	Key Features	Strengths	Weaknesses	Market Positioning
Praava Health	Online appointment booking, telemedicine, diagnostics	Strong urban presence, high-quality care, digital channels	High cost, urban-centric, limited offline reach	Urban middle/upper-class, premium
Tonic / Grameenphone	Health hotline, mobile consults, MFS integration	Large telco user base, simple access, inclusive	Limited advanced features, less focus on chronic care & e-prescriptions	Mass market, value-based
Maya	Symptom checker, multi-language chat, women-centric care	AI-powered triage, intuitive chat UI	No pharmacy/e-prescription network, limited voice/video options	Young urban users, mobile-first
Doctorola	Doctor search, booking, video consults	Broad doctor network, multiple channels	No offline onboarding, higher digital literacy required, weak in MFS rewards	Appointment aggregator, urban focus

2.3 User Needs and Pain Points

User Need / Pain Point	Observed Behavior / Quote	Proposed Solution
Difficulty accessing reliable care in rural areas	“I rely on local health worker’s advice, I don’t trust apps.”	PHC worker-driven offline onboarding, community trust building
App usability anxiety due to language and complex tech	“I only understand Bangla, and apps are confusing.”	Multilingual UI, simplified workflows
Poor internet connectivity breaks consultations	“Video calls don’t work with my network.”	Data-light modes, voice-first options
High out-of-pocket medicine costs, fear of counterfeit drugs	“Medicine is expensive, sometimes not real.”	e-Prescription, pharmacy network with cashback
Payment distrust; unfamiliarity with cards or banks	“I only use bKash, don’t have a bank card.”	MFS integration with bKash/Nagad
Lack of reminders for ongoing or chronic care	“I forget my follow-up and meds.”	SMS reminders, chronic disease modules
No financial safety net for hospitalizations	“Hospital bills are impossible to pay.”	Hospitalization cashback via MFS

3.0 User Stories

3.1 User Personas

Persona 1:

- **Name & Photo:** Ayesha Begum (Stock photo: Rural Bangladeshi woman, sari, in home courtyard)
- **Demographics:** 38, Homemaker, rural Comilla
- **Bio:** Manages family’s daily health, relies on local health worker and small pharmacy, limited education, uses basic feature phone.
- **Goals:** Timely medical advice, affordable medicines, care for elders.
- **Frustrations:** Long travel to clinics, high costs, distrust in digital tools, fear of fake medicines.
- **Motivations:** Community approval, saving money, peace of mind for family.

Persona 2:

- **Name & Photo:** Rafiqul Islam (Stock photo: Male, 29, motorcycle, in town marketplace)

- **Demographics:** 29, Small business owner, peri-urban Bogura
- **Bio:** Runs local grocery store, moderate smartphone user, busy but health-conscious, has parents with chronic diseases.
- **Goals:** Fast consultations, manage parents' medications, avoid work disruptions.
- **Frustrations:** Internet unreliability, difficulty in app navigation, managing chronic prescriptions.
- **Motivations:** Convenience, reliability, technology that "just works."

Persona 3:

- **Name & Photo:** Salma Akter (Stock photo: Primary health worker, 46, in PHC clinic uniform)
- **Demographics:** 46, PHC worker, rural Rajshahi
- **Bio:** Trained in public health, trusted in her village, acts as a bridge between community and formal health system.
- **Goals:** Provide timely referrals, support families, improve health outcomes.
- **Frustrations:** Limited tools for patient follow-up, difficulty explaining digital solutions.
- **Motivations:** Community respect, professional pride, tools that empower her work.

3.2 User Journey Maps

Scenario: Ayesha Begum's Healthcare Journey

Discovery:

- Action: Learns about the digital health service via local PHC worker during a village meeting.
- Thoughts: "Will this really help my family? Can I trust it?"
- Feelings: Curious, cautious, hopeful.

Consideration:

- Action: Sees a demonstration, asks questions in Bangla, PHC worker reassures her.
- Thoughts: "If she recommends it, maybe it's good. Is it hard to use?"
- Feelings: Reassured by community trust.

Onboarding:

- Action: Downloads app via local WiFi, guided step-by-step in Bangla by PHC worker.
- Thoughts: "This isn't as hard as I thought. It's in my language."
- Feelings: Empowered, less anxious.

Usage:

- Action: Checks symptoms with simple checker, connects via voice (not video), pays with bKash, receives e-prescription.
- Thoughts: “No need to travel. Paid with bKash, got medicine easily.”
- Feelings: Relieved, grateful.

Advocacy:

- Action: Shares experience at women’s meeting, encourages neighbors to try.
- Thoughts: “Others should use this too—it’s safe and affordable.”
- Feelings: Proud, confident.

4.0 Features

4.1 Feature Prioritization

Feature Name	Priority (MoSCoW)	Justification
Offline Marketing & PHC Training	Must-have	Essential for onboarding rural and tech-averse users
Low-Bandwidth App Download	Must-have	Enables reach where connectivity is poor
Multilingual UI (Bangla/English)	Must-have	Reduces technology anxiety and encourages broad adoption
Symptom Checker (Simple)	Must-have	Facilitates digital triage for all users
MFS Integration (bKash/Nagad)	Must-have	Familiar payment channels are critical for trust and adoption
Voice/Video Consultation	Must-have	Offers flexible options for users based on bandwidth and comfort
Data-Light Consultation Modes	Should-have	Improves reliability for poor networks
Digital Patient History	Should-have	Continuity of care and chronic disease management
e-Prescription	Must-have	Ensures medicine authenticity and streamlines post-consult experience
Pharmacy Network Integration	Should-have	Drives redemption, builds trust in prescribed medicines

MFS Cashback	Should-have	Incentivizes affordable care and medicine adherence
SMS Reminders	Should-have	Boosts ongoing care and reduces missed follow-ups
Chronic Disease Management	Could-have	Enhances long-term health outcomes, important for future phases
Hospitalization Cashback	Could-have	Adds critical financial safety net for severe cases

4.2 Feature Dependencies

- Low-Bandwidth App Download depends on Multilingual UI being implemented for accessibility.
- Voice/Video Consultation depends on MFS Integration for seamless payment before session start.
- e-Prescription requires Digital Patient History for continuity and accuracy of records.
- Pharmacy Network Integration requires e-Prescription functionality to be in place.
- SMS Reminders depend on Digital Patient History for timely, personalized notifications.
- Hospitalization Cashback depends on successful MFS Integration and user verification.

4.3 Feature Acceptance Criteria

Feature: Multilingual UI (Bangla/English)

Scenario 1: Successful Language Selection

Given the user installs the app for the first time
When they reach the language selection screen
Then they can choose Bangla or English and the entire UI updates accordingly

Scenario 2: Switching Language Post-Onboarding

Given a logged-in user is on the dashboard
When they navigate to 'Settings' and select a different language
Then the UI immediately updates to the chosen language and all subsequent screens are displayed accordingly

Scenario 3: Error Handling for Unsupported Language

Given the user attempts to select an unsupported language
When the selection is made
Then the system displays a clear error message and defaults to Bangla

5.0 Technical Requirements

5.1 Data Sources

- Internal user database

- Pharmacy network directory API
- MFS (bKash/Nagad) transaction APIs
- User-generated health history
- Third-party SMS gateway

Sample Data Object: Digital Patient History (JSON)

```
{
  "patient_id": "U123456",
  "name": "Ayesha Begum",
  "dob": "1986-04-12",
  "language": "bn",
  "consultations": [
    {
      "date": "2024-06-10",
      "symptoms": ["headache", "fever"],
      "doctor_id": "D9876",
      "consult_type": "voice",
      "prescription_id": "RX12345"
    }
  ],
  "chronic_conditions": ["hypertension"],
  // Stores only non-PII for reporting and reminders
}
```

5.2 System Architecture

The proposed platform will employ a cloud-native architecture, using React Native for the mobile front end to ensure a seamless, cross-platform experience. The backend will be powered by Node.js and Express for flexible API management, connected to a PostgreSQL database for secure, scalable data storage. Core services will be containerized via Docker and orchestrated using Kubernetes, deployed on AWS for global availability and auto-scaling. This stack enables rapid iteration, robustness under variable loads, and aligns with the team’s expertise in agile cloud deployments and local language support.

5.3 Performance Benchmarks

- API endpoints must respond in under 250ms under normal load
- App must load on 2G networks in under 5 seconds
- Largest Contentful Paint (LCP) under 2.5s for all primary screens
- System should support 5,000 concurrent users without degradation
- 99.9% uptime with automated failover and recovery

5.4 Security and Privacy

- All data encrypted at rest and in transit (TLS 1.2+)
- Strict role-based access controls for PHC workers vs. end users
- PII anonymized for analytics and reporting
- GDPR and local data privacy regulations adhered to
- Secure MFS integration using tokenized credentials

Sample SQL for Anonymized User Data Retrieval:

```
SELECT
  SUBSTRING(name, 1, 1) || '***' AS anonymized_name,
  YEAR(dob) AS birth_year,
  language,
  chronic_conditions
FROM users
WHERE consent_for_research = TRUE;
```

6.0 Design and User Experience

6.1 Information Architecture

- Dashboard
    - Symptom Checker
    - Book Consultation
        - Voice Call
        - Video Call
    - Digital Patient History
        - Previous Consultations
        - Chronic Disease Management
    - e-Prescriptions
        - View/Download
        - Pharmacy Network Map
    - Payments
        - MFS Integration
        - Cashback Offers
    - Settings
        - Language Selection
        - Notifications (SMS Reminders)
        - Profile

6.2 Wireframes or Mockups

Consultation Booking Screen (Voice/Video)

- Header: “Book a Consultation” with language toggle (Bangla/English) on top right.
- Step 1: Select Consultation Type
    - Two prominent buttons: “Voice Call” and “Video Call,” with clear icons
- Step 2: Symptom Input Field

    - Single row, large font input for symptoms, with auto-complete suggestions in selected language
- Step 3: Doctor Preferences
    - Dropdown (optional): “Any Doctor” or select from specialties
- Step 4: Payment Selection
    - bKash/Nagad icon buttons with short instructional text
- Step 5: Confirm & Pay Button
    - Primary action button, disabled until payment info is entered
- Footer: Customer support hotline, SMS reminder opt-in checkbox

Interactive Behavior:

- All fields validate and provide feedback in real time
- Language toggle updates all text and labels instantly
- Payment errors shown in red with suggested resolution

6.3 Usability Testing Plan

Testing Phase	Methodology	Participants	Key Questions to Answer
Early Wireframes	Moderated interviews, think-aloud	10 rural users, 5 PHC	Can users navigate onboarding? Is language clear?
Pre-launch Beta	Unmoderated task analysis	50 active users	Can all core flows (consult, pay, redeem) be completed without help? What errors occur?

7.0 Project Management

7.1 Project Timeline

Phase	Key Milestones	Estimated Duration (Weeks)
Discovery & Planning	User research, requirements gathering	2
Design & Prototyping	Wireframes, interactive prototypes	3
Development	Onboarding, low-bandwidth app,	3

Sprint 1	MFS integration	
Development Sprint 2	Consultation, symptom checker, e-prescription	3
Testing & QA	Usability, security, performance validation	2
Launch	Marketing rollout, PHC training, app release	1

7.2 Resource Allocation

Role	Key Responsibilities
Product Manager	Roadmap, requirements, stakeholder alignment
Lead Engineer	System architecture, backend/frontend supervision
Mobile Developer	App development, offline/low-bandwidth optimizations
UI/UX Designer	Wireframes, language localization, usability enhancements
QA Analyst	Functional testing, performance, data privacy verification
PHC Training Lead	Train-the-trainer, community onboarding coordination
Marketing Specialist	Offline campaigns, PHC worker support, rural outreach

7.3 Risk Management

Risk Description	Likelihood	Impact	Mitigation Strategy
Low digital literacy delays user adoption	High	High	Comprehensive PHC-led onboarding and ongoing support
Network outages or poor connectivity	Medium	High	Data-light modes, robust offline support
Regulatory change or new digital health compliance rules	Medium	Medium	Continuous monitoring, agile legal/tech team alignment
MFS integration or payout delays	Low	High	Multiple payment partners, fallback workflows
User mistrust in digital	High	Medium	Community trust campaigns,

health platforms			transparent data handling

7.4 Communication Plan

Communication Type	Audience	Channel	Frequency
Daily Standup	Core team	Slack/Meeting	Daily
Weekly Stakeholder Update	Internal stakeholders	Email/Meeting	Weekly
PHC Worker Sync	PHC leads, trainers	Phone/WhatsApp	Bi-weekly
Progress Review	Leadership/Investors	Presentation	Monthly
User Feedback Roundup	All teams	Shared dashboard	Fortnightly

8.0 Appendices

8.1 Supporting Documentation

N/A (No supporting documents provided)Car